Linear Algebra and Its Applications

Section 3.2: Properties of determinants

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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3.2 Properties of determinants

A row replacement preserves a determinant, a row interchange reverses its sign, and multiplying a single row by k multiplies it by k. In the calculations below, keeping the operations visible is how we avoid losing a sign or a scale factor. Once a matrix is triangular its determinant is the product of its diagonal entries.

A square matrix is invertible exactly when its determinant is nonzero. Thus n vectors in $\mathbb{R}^n$ are linearly independent exactly when the determinant of the matrix having those vectors as columns is nonzero. The multiplication theorem says $\det(AB) = \det A \det B$, and transposition preserves a determinant.

Practice problem 3.2.P1: Create a zero row

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_4 \leftarrow R_4 + 3R_1$. Both resulting rows 2 and 4 are $(0, 1, -3, 2)$. Then $R_4 \leftarrow R_4 - R_2$ creates a zero row. All three operations preserve the determinant, and expansion along the zero row gives $\det A = 0$.

Practice problem 3.2.P2: A determinant tests independence

Place the three vectors in columns, not rows:

$$A = \begin{bmatrix} 5 & -3 & 2 \\ -7 & 3 & -7 \\ 9 & -5 & 5 \end{bmatrix}.$$

Replace R_2 by $R_2 + R_1 = (-2, 0, -5)$. Expanding down column 2 now gives

$$\det A = 3\begin{vmatrix} -2 & -5 \\ 9 & 5 \end{vmatrix} + 5\begin{vmatrix} 5 & 2 \\ -2 & -5 \end{vmatrix} = 3(35) + 5(-21) = 0.$$

Thus A is not invertible, and the three columns are linearly dependent by the Invertible Matrix Theorem.

Exercise 3.2.1: Identify the determinant rule

The displayed matrices differ by $R_1 \leftrightarrow R_2$. A single interchange reverses the determinant's sign, which explains the minus sign in the equation.

Exercise 3.2.2: Identify the determinant rule

The first row $(2, -6, 4)$ is $2(1, -3, 2)$, while the other rows are fixed. Linearity in one row gives exactly the displayed factor 2.

Exercise 3.2.3: Identify the determinant rule

The right-hand matrix results from $R_2 \leftarrow R_2 - 2R_1$, since $(2, 0, -3) - 2(1, 3, -4) = (0, -6, 5)$. A row replacement preserves the determinant.

Exercise 3.2.4: Identify the determinant rule

Apply $R_3 \leftarrow R_3 - 3R_1$: $(3,7,4) - 3(1,2,3) = (0,1,-5)$. This is a row replacement and does not change the determinant.

Exercise 3.2.5: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & -6 \\ -1 & -4 & 4 \\ -2 & -7 & 9 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-1)R_1, \quad R_3 \leftarrow R_3 - (-2)R_1$, giving

$$\begin{bmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 3 & -3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (3)R_2$, giving

$$\begin{bmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 0 & 3 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(3) = 3.$$

Exercise 3.2.6: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & -3 \\ 3 & -3 & 3 \\ 2 & 13 & -7 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (3)R_1, \quad R_3 \leftarrow R_3 - (2)R_1$, giving

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & -18 & 12 \\ 0 & 3 & -1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{1}{6})R_2$, giving

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & -18 & 12 \\ 0 & 0 & 1 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(-18)(1) = -18.$$

Exercise 3.2.7: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-2)R_1, \quad R_3 \leftarrow R_3 - (3)R_1, \quad R_4 \leftarrow R_4 - (1)R_1$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & -4 & 2 & -5 \\ 0 & -4 & 2 & -5 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-4)R_2, \quad R_4 \leftarrow R_4 - (-4)R_2$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & 0 & 30 & 27 \\ 0 & 0 & 30 & 27 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (1)R_3$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & 0 & 30 & 27 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(30)(0) = 0.$$

Exercise 3.2.8: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 2 & 5 & 4 & -3 \\ -3 & -7 & -5 & 2 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3 - (2)R_1, \quad R_4 \leftarrow R_4 - (-3)R_1$, giving

$$\begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 0 & -1 & -2 & 5 \\ 0 & 2 & 4 & -10 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-1)R_2, \quad R_4 \leftarrow R_4 - (2)R_2$, giving

$$\begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(0)(0) = 0.$$

Exercise 3.2.9: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ -1 & 2 & 8 & 5 \\ 3 & -1 & -2 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3 - (-1)R_1, \quad R_4 \leftarrow R_4 - (3)R_1$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 1 & 5 & 5 \\ 0 & 2 & 7 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (1)R_2, \quad R_4 \leftarrow R_4 - (2)R_2$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -3 & -5 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & -3 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(1)(1)(-3)(1) = 3.$$

Exercise 3.2.10: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ -2 & -6 & 2 & 3 & 9 \\ 3 & 7 & -3 & 8 & -7 \\ 3 & 5 & 5 & 2 & 7 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3-(-2)R_1, \quad R_4 \leftarrow R_4-(3)R_1, \quad R_5 \leftarrow R_5-(3)R_1$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & -2 & 0 & 8 & -1 \\ 0 & -4 & 8 & 2 & 13 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4-(-1)R_2, \quad R_5 \leftarrow R_5-(-2)R_2$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & 0 & -4 & 7 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & -4 & 7 & -7 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(1)(2)(-4)(3)(1) = 24.$$

Exercise 3.2.11: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2-(\frac{3}{2})R_1, \quad R_3 \leftarrow R_3-(-3)R_1, \quad R_4 \leftarrow R_4-(2)R_1$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 15 & -13 & 6 \\ 0 & 0 & 2 & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3-(-2)R_2$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 2 & 1 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-1)R_3$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-\frac{15}{2})(-2)(4) = 120.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.12: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ 5 & 4 & 6 & 6 \\ 4 & 2 & 4 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-3)R_1, \quad R_3 \leftarrow R_3 - (-5)R_1, \quad R_4 \leftarrow R_4 - (-4)R_1$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 14 & 21 & 6 \\ 0 & 10 & 16 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{7}{5})R_2, \quad R_4 \leftarrow R_4 - (1)R_2$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 0 & \frac{21}{5} & 6 \\ 0 & 0 & 4 & 3 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (\frac{20}{21})R_3$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 0 & \frac{21}{5} & 6 \\ 0 & 0 & 0 & -\frac{19}{7} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(-1)(10)(\frac{21}{5})(-\frac{19}{7}) = 114.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.13: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 2 & 5 & 4 & 1 \\ 4 & 7 & 6 & 2 \\ 6 & -2 & -4 & 0 \\ -6 & 7 & 7 & 0 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (2)R_1, \quad R_3 \leftarrow R_3 - (3)R_1, \quad R_4 \leftarrow R_4 - (-3)R_1$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & -17 & -16 & -3 \\ 0 & 22 & 19 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{17}{3})R_2, \quad R_4 \leftarrow R_4 - (-\frac{22}{3})R_2$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & 0 & -\frac{14}{3} & -3 \\ 0 & 0 & \frac{13}{3} & 3 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-\frac{13}{14})R_3$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & 0 & -\frac{14}{3} & -3 \\ 0 & 0 & 0 & \frac{3}{14} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-3)(-\frac{14}{3})(\frac{3}{14}) = 6.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.14: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} -3 & -2 & 1 & -4 \\ 1 & 3 & 0 & -3 \\ -3 & 4 & -2 & 8 \\ 3 & -4 & 0 & 4 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-\frac{1}{3})R_1, \quad R_3 \leftarrow R_3 - (1)R_1, \quad R_4 \leftarrow R_4 - (-1)R_1$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 6 & -3 & 12 \\ 0 & -6 & 1 & 0 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{18}{7})R_2, \quad R_4 \leftarrow R_4 - (-\frac{18}{7})R_2$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 0 & -\frac{27}{7} & \frac{162}{7} \\ 0 & 0 & \frac{13}{7} & -\frac{78}{7} \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-\frac{13}{27})R_3$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 0 & -\frac{27}{7} & \frac{162}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(-3)(\frac{7}{3})(-\frac{27}{7})(0) = 0.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.15: Transform a known determinant

The third row is scaled by 5 and the other rows are fixed. Thus the determinant is $5 \cdot 7 = 35$.

Exercise 3.2.16: Transform a known determinant

The second row is scaled by 3, so the determinant is $3 \cdot 7 = 21$.

Exercise 3.2.17: Transform a known determinant

Interchanging rows 2 and 3 once gives the displayed matrix. Its determinant is -7.

Exercise 3.2.18: Transform a known determinant

Move the original third row to the top by two interchanges: first interchange rows 2 and 3, then rows 1 and 2. The two sign changes cancel, leaving determinant 7.

Exercise 3.2.19: Transform a known determinant

First scale the second row by 2, multiplying the determinant by 2. Then add row 1 to it, which preserves that new determinant. The answer is $2 \cdot 7 = 14$.

Exercise 3.2.20: Transform a known determinant

The operation $R_1 \leftarrow R_1 + R_2$ preserves the determinant, so the answer is 7.

Exercise 3.2.21: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 2 & 3 & 0 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{2})R_1, \quad R_3 \leftarrow R_3 - (\frac{1}{2})R_1$, giving

$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & \frac{3}{2} & 4 \\ 0 & \frac{1}{2} & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{1}{3})R_2$, giving

$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & \frac{3}{2} & 4 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(\frac{3}{2})(-\frac{1}{3}) = -1.$$

Since the determinant is -1, the matrix is invertible by the determinant criterion.

Exercise 3.2.22: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 5 & 0 & -1 \\ 1 & -3 & -2 \\ 0 & 5 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{5})R_1$, giving

$$\begin{bmatrix} 5 & 0 & -1 \\ 0 & -3 & -\frac{9}{5} \\ 0 & 5 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{5}{3})R_2$, giving

$$\begin{bmatrix} 5 & 0 & -1 \\ 0 & -3 & -\frac{9}{5} \\ 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(5)(-3)(0) = 0.$$

Since the determinant is 0, the matrix is not invertible by the determinant criterion.

Exercise 3.2.23: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 2 & 0 & 0 & 8 \\ 1 & -7 & -5 & 0 \\ 3 & 8 & 6 & 0 \\ 0 & 7 & 5 & 4 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{2})R_1, \quad R_3 \leftarrow R_3 - (\frac{3}{2})R_1$, giving

$$\begin{bmatrix} 2 & 0 & 0 & 8 \\ 0 & -7 & -5 & -4 \\ 0 & 8 & 6 & -12 \\ 0 & 7 & 5 & 4 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{8}{7})R_2, \quad R_4 \leftarrow R_4 - (-1)R_2$, giving

$$\begin{bmatrix} 2 & 0 & 0 & 8 \\ 0 & -7 & -5 & -4 \\ 0 & 0 & \frac{2}{7} & -\frac{116}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-7)(\frac{2}{7})(0) = 0.$$

Since the determinant is 0, the matrix is not invertible by the determinant criterion.

Exercise 3.2.24: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 4 & -7 & -3 \\ 6 & 0 & -5 \\ -7 & 2 & 6 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{3}{2})R_1, \quad R_3 \leftarrow R_3 - (-\frac{7}{4})R_1$, giving

$$\begin{bmatrix} 4 & -7 & -3 \\ 0 & \frac{21}{2} & -\frac{1}{2} \\ 0 & -\frac{41}{4} & \frac{3}{4} \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{41}{42})R_2$, giving

$$\begin{bmatrix} 4 & -7 & -3 \\ 0 & \frac{21}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{11}{42} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(4)(\frac{21}{2})(\frac{11}{42}) = 11.$$

There are 3 vectors in $\mathbb{R}^3$, so the determinant test applies. The determinant is 11; hence the set is linearly independent.

Exercise 3.2.25: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 7 & -8 & 7 \\ -4 & 5 & 0 \\ -6 & 7 & -5 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-\frac{4}{7})R_1, \quad R_3 \leftarrow R_3 - (-\frac{6}{7})R_1$, giving

$$\begin{bmatrix} 7 & -8 & 7 \\ 0 & \frac{3}{7} & 4 \\ 0 & \frac{1}{7} & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{1}{3})R_2$, giving

$$\begin{bmatrix} 7 & -8 & 7 \\ 0 & \frac{3}{7} & 4 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(7)(\frac{3}{7})(-\frac{1}{3}) = -1.$$

There are 3 vectors in $\mathbb{R}^3$, so the determinant test applies. The determinant is -1; hence the set is linearly independent.

Exercise 3.2.26: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 3 & 2 & -2 & 0 \\ 5 & -6 & -1 & 0 \\ -6 & 0 & 3 & 0 \\ 4 & 7 & 0 & -3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{5}{3})R_1, \quad R_3 \leftarrow R_3 - (-2)R_1, \quad R_4 \leftarrow R_4 - (\frac{4}{3})R_1$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 4 & -1 & 0 \\ 0 & \frac{13}{3} & \frac{8}{3} & -3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{3}{7})R_2, \quad R_4 \leftarrow R_4 - (-\frac{13}{28})R_2$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{15}{4} & -3 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 0 & \frac{15}{4} & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(3)(-\frac{28}{3})(\frac{15}{4})(0) = 0.$$

There are 4 vectors in $\mathbb{R}^4$, so the determinant test applies. The determinant is 0; hence the set is linearly dependent.

Exercise 3.2.27: Properties, proofs, and exact calculations

(a) True.

This is the row-replacement part of Theorem 3.

(b) False as stated for an arbitrary echelon form.

The proposed sign correction accounts for row interchanges, but not row scalings. Starting with $A = I_2$, scale row 1 by 2. The resulting echelon form has pivot product 2 and no interchanges, although $\det A = 1$. If reduction uses *only* replacements and interchanges, the stated product with its sign correction is valid.

(c) True.

Dependent columns mean that A is not invertible, by the Invertible Matrix Theorem; Theorem 4 then gives $\det A = 0$.

(d) False.

Take $A = B = I_2$. The left side is $\det(2I_2) = 4$, while the right side is $1 + 1 = 2$.

Exercise 3.2.28: Properties, proofs, and exact calculations

(a) True.

Each interchange multiplies the determinant by -1, so two multiply it by $(-1)^2 = 1$.

(b) False.

The diagonal-product rule needs a triangular matrix. For example, $\left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ has determinant 0, but diagonal product 1.

(c) False.

The matrix $\left[\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{smallmatrix}\right]$ has row 3 equal to row 1 plus row 2 and therefore determinant zero. None of its rows or columns is zero or identical to another.

(d) False.

Theorem 5 gives $\det A^T = \det A$. The identity matrix, whose determinant is 1, disproves the proposed minus sign.

Exercise 3.2.29: Properties, proofs, and exact calculations

For $B = \left[\begin{smallmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{smallmatrix}\right]$, subtract row 1 from rows 2 and 3, giving lower right block $\left[\begin{smallmatrix} 1 & 1 \\ 2 & 0 \end{smallmatrix}\right]$. Thus $\det B = 1(0 - 2) = -2$. Applying multiplicativity repeatedly gives $\det(B^5) = (\det B)^5 = (-2)^5 = -32$.

Exercise 3.2.30: Properties, proofs, and exact calculations

Interchange the two equal rows. The numerical matrix does not change, but Theorem 3 says its determinant changes sign. Thus $d = -d$, where $d = \det A$. Adding d gives $2d = 0$, hence $d = 0$. If two columns are equal, transpose the matrix; its transpose has two equal rows and the same determinant by Theorem 5. This proof does not use the invertibility criterion in Theorem 4.

Exercise 3.2.31: Properties, proofs, and exact calculations

Since A is invertible, Theorem 4 gives $\det A \neq 0$. The multiplication theorem applied to $AA^{-1} = I$ gives $(\det A)(\det A^{-1}) = \det I = 1$. Dividing by the nonzero scalar $\det A$ proves $\det A^{-1} = 1/\det A$.

Exercise 3.2.32: Properties, proofs, and exact calculations

The matrix rA is obtained by multiplying each of the n rows by r. Repeated use of Theorem 3(c), one row at a time, gives $\det(rA) = r^n \det A$. If $r = 0$ and $n \geq 1$, the matrix is zero and both sides are zero, so that case is also covered.

Exercise 3.2.33: Properties, proofs, and exact calculations

Theorem 6 and commutativity of *real-number* multiplication give $\det(AB) = (\det A)(\det B) = (\det B)(\det A) = \det(BA)$. This does not assert that the matrix products themselves commute.

Exercise 3.2.34: Properties, proofs, and exact calculations

By multiplicativity, $\det(PAP^{-1}) = (\det P)(\det A)(\det P^{-1})$. Because P is invertible, Exercise 31 gives $(\det P)(\det P^{-1}) = 1$. The product therefore reduces to $\det A$. Similar matrices have the same determinant.

Exercise 3.2.35: Properties, proofs, and exact calculations

Take determinants of $U^TU = I$. Theorems 5 and 6 give $(\det U)^2 = (\det U^T)(\det U) = 1$. Since $\det U$ is real, solving $d^2 = 1$ gives $d = 1$ or $d = -1$. Both signs occur: I gives 1, and a single coordinate reflection gives -1.

Exercise 3.2.36: Properties, proofs, and exact calculations

Repeated multiplication gives $0 = \det(A^4) = (\det A)^4$. A real number has fourth power zero only when it is zero. Thus $\det A = 0$, so Theorem 4 rules out invertibility.

Exercise 3.2.37: Properties, proofs, and exact calculations

Multiply the matrices first:

$$AB = \begin{bmatrix} 6 & 0 \\ 17 & 4 \end{bmatrix}.$$

Direct 2×2 calculations give $\det A = 3$, $\det B = 8$, and

$$\det(AB) = (6)(4) - (0)(17) = 24 = (\det A)(\det B).$$

This is a direct verification for these matrices, rather than an application of Theorem 6.

Exercise 3.2.38: Properties, proofs, and exact calculations

Multiply the matrices first:

$$AB = \begin{bmatrix} 6 & 0 \\ -2 & 0 \end{bmatrix}.$$

Direct 2×2 calculations give $\det A = 0$, $\det B = -2$, and

$$\det(AB) = (6)(0) - (0)(-2) = 0 = (\det A)(\det B).$$

This is a direct verification for these matrices, rather than an application of Theorem 6.

Exercise 3.2.39: Properties, proofs, and exact calculations

Use the given 3×3 dimension when scaling every row:

(a) $\det(AB) = 4(-3) = -12$, by multiplicativity.

(b) $\det(5A) = 5^3 \cdot 4 = 500$, since all three rows are scaled.

(c) $\det(B^T) = \det B = -3$, by transposition invariance.

(d) $\det(A^{-1}) = 1/4$, since $\det A \neq 0$.

(e) $\det(A^3) = 4^3 = 64$, by applying multiplicativity twice.

Exercise 3.2.40: Properties, proofs, and exact calculations

The matrices are now 4×4:

(a) $\det(AB) = (-1)(2) = -2$.

(b) $\det(B^5) = 2^5 = 32$.

(c) $\det(2A) = 2^4(-1) = -16$.

(d) $\det(A^T A) = \det(A^T) \det A = (-1)^2 = 1$.

(e) $\det(B^{-1}AB) = (1/2)(-1)(2) = -1$; the inverse exists because $\det B = 2 \neq 0$.

Exercise 3.2.41: Properties, proofs, and exact calculations

Expand each 2×2 determinant directly:

$$\det A = (a + e)d - (b + f)c = (ad - bc) + (ed - fc) = \det B + \det C.$$

Only the *first row* was split, while the second row stayed fixed. This is compatible with failure of additivity in the whole matrix.

Exercise 3.2.42: Properties, proofs, and exact calculations

Here $A = I_2$. Direct calculation yields

$$\det(I_2 + B) = (1 + a)(1 + d) - bc = 1 + ad - bc + (a + d).$$

But $\det I_2 + \det B = 1 + ad - bc$. The two expressions are equal if and only if their difference $a + d$ is zero. This proves both implications of the claimed equivalence.

Exercise 3.2.43: Properties, proofs, and exact calculations

Expand each determinant down its third column. The cofactors C_{13}, C_{23}, C_{33} depend only on the first two columns, so they are the same for all three matrices. Therefore

$$\begin{aligned}\det A &= \sum_{i=1}^{3}(u_i + v_i)C_{i3} \\ &= \sum_{i=1}^{3} u_i C_{i3} + \sum_{i=1}^{3} v_i C_{i3} = \det B + \det C.\end{aligned}$$

The matrices do not satisfy $A = B + C$, because adding B and C would double their first two columns. The equality is linearity in one column with the other columns held fixed.

Exercise 3.2.44: Properties, proofs, and exact calculations

Transpose first: $(AE)^T = E^T A^T$. The transpose E^T is an elementary matrix of the corresponding type. Theorem 3, applied to left multiplication by E^T, gives

$$\det(E^T A^T) = (\det E^T)(\det A^T).$$

By Theorem 5, the left side equals $\det(AE)$, and the right side equals $(\det E)(\det A)$. This argument uses row-operation rules and transpose invariance, not the general product theorem.

Exercise 3.2.45: Rectangular matrices and two Gram products

Run the following exact experiment, including two samples at each requested size.

```
from random import Random
from sympy import Matrix
r = Random(32045)
for m,n in ((4,5),(4,5),(5,6),(5,6)):
    A = Matrix(m,n,[r.randint(-5,5) for _ in range(m*n)])
    print(m,n,(A.T*A).det(),(A*A.T).det())
```

The computed outputs are

shape of A	$\det(A^T A)$	$\det(AA^T)$
4×5	0	22645
4×5	0	128772
5×6	0	33391308
5×6	0	1276024

Explain what must hold for every such matrix.

If A has $m < n$, its n columns in $\mathbb{R}^m$ are dependent, so some nonzero x satisfies $Ax = 0$. Then $A^T Ax = 0$, proving that $A^T A$ is singular and its determinant is zero. In contrast, AA^T is invertible exactly when the rows of A are independent: for any y,

$$y^T AA^T y = \|A^T y\|^2.$$

This is zero exactly when $A^T y = 0$. Thus independent rows imply a zero null space for AA^T. A dependent-row sample would also give $\det(AA^T) = 0$; its nonzero values in these four experiments are not a universal guarantee.

Exercise 3.2.46: Small determinant versus near singularity

What we must show. Compare scaling-sensitive determinants with a scaling-invariant condition number. The specified matrix from Section 2.3, Exercise 9 is

$$A = \begin{bmatrix} 4 & 0 & -3 & -7 \\ -6 & 9 & 9 & 9 \\ 7 & -5 & 10 & 19 \\ -1 & 2 & 4 & -1 \end{bmatrix}.$$

Using the spectral (2-norm) condition number, the following double-precision calculation gives the table below.

```
import numpy as np
A = np.array([[4,0,-3,-7],[-6,9,9,9],
              [7,-5,10,19],[-1,2,4,-1]],float)
for B in (A,np.eye(4)):
    for c in (0.1,1.,10.):
        print(c,np.linalg.det(c*B),np.linalg.cond(c*B,2))
```

The determinants in the table were additionally checked in exact rational arithmetic, avoiding cancellation errors in their reported values.

matrix	det	κ_2
$\frac{1}{10}A$	$-\frac{501}{1250}$	11.3525
$1A$	-4008	11.3525
$10A$	-40080000	11.3525
$\frac{1}{10}I_4$	$\frac{1}{10000}$	1
$1I_4$	1	1
$10I_4$	10000	1

Interpret the comparison.

The supplied A is less well-conditioned than the identity, with a moderate condition number about 11.35. The exercise calls it nearly singular, but the actual supplied entries do not exhibit severe ill-conditioning. Its determinant changes by a factor 10^4 when A is multiplied by 10, while its condition number does not change. In general, $\det(cA) = c^4 \det A$ in this dimension. But for nonzero c,

$$\kappa_2(cA) = \|cA\|_2 \|(cA)^{-1}\|_2 = |c| \|A\|_2 \, |c|^{-1} \|A^{-1}\|_2 = \kappa_2(A).$$

Every nonzero scalar multiple of the identity has condition number 1, even if its determinant is made arbitrarily small. Thus determinant size alone, without controlling matrix scale, does not measure relative nearness to singularity. Exact zero still characterizes singularity; "small" is the scale-dependent part.